THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

December 20th, 2024 (Friday) 9:30 AM – 12:30 PM (3 hours)

Answer ALL **5 (FIVE)** questions from Section A and B.
Marks for sub-questions (if any) are given.

Use a separate answer book for Section A and Section B.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

Q1 (a) As shown in Figure 1(a), the beam ABCD is subject to external moment at B and C and concentrated load at mid-point of CD span. Distance from A to C is L. Distance from C to D is 0.5L.

(i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

(ii) Draw the bending moment diagram (BDM) with pertinent values. Noe that bending moment should be drawn on the **tensile** side. **(5 marks)**

(b) A frame is as shown in Figure 1(b). Support A and B are both pin supports. Connection C is a hinge connection. Total height is L and distance from A to B is 2L. The loading is as shown in Figure 1(b).

(i) Draw the shear force diagram (SFD) of the frame with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. Note that SFD of all components need to be plotted. **(5 marks)**

(ii) Draw the bending moment diagram (BDM) of the frame with pertinent values. Noe that bending moment should be drawn on the **tensile** side. Note that BMD of all components need to be plotted. **(5 marks)**

Q2 (a) A cantilever metal beam in Figure 2(a) has the section as shown in Figure 2(b). The total length is L=10m. The load is q=20kN/m. The web thickness is 4 mm, the height of web is 500 mm. The top and bottom flange plates are 300 mm width and 25 mm thick. Only consider external loading of the steel beam. Neglect buckling for this beam. The von Mises criterion is used for metal material. The yield strength is obtained from tensile test as 300 MPa. The safety factor is 1.0.

(i) Check the maximum normal stress and maximum shear stress of the beam. **(5 marks)**

(ii) Check the safety of the beam using von Mises criterion. **(5 marks)**

(b) A point is subject to plane stress state as shown in Figure 2(c). x axis is horizontal and pointing to the right. y axis is vertical pointing upward. The von Mises criterion is used for the material. The yield strength is obtained from tensile test as 200 MPa. Safety factor is 1.0.

(i) Determine the stress acting on an element oriented at an angle θ=45 degrees from the x axis. Rotating counterclockwise from x axis is positive. **(5 marks)**

(ii) Check the safety of the point using von Mises criterion. **(5 marks)**

Q3 A long slender column ABC is pinned at ends A and C and compressed by an axial force P. At midpoint B, lateral support is provided to prevent deflection in the plane of the figure. All bars are connected by hinge connection. Each member is a slender steel bar with elastic modulus of 200 GPa. The cross section of the members is as shown in Figure 3(b). The distance between points A and B is L=30 m. Assume the material will not fail due to strength issue. For buckling design, safety factor of 2 is needed for this problem. Consider the possibility of Euler buckling about either principal centroidal axis (i.e. strong axis or weak axis).

(i) Determine the allowable value of the load *P*. **(8 marks)**

(ii) When allowable value *P* is reached, determine the displacement at point C. **(2 marks)**

FIGURES

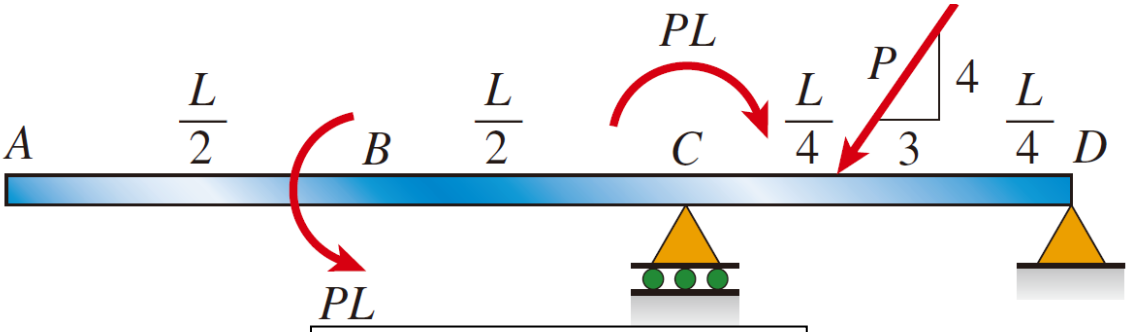


Figure 1(a). Beam

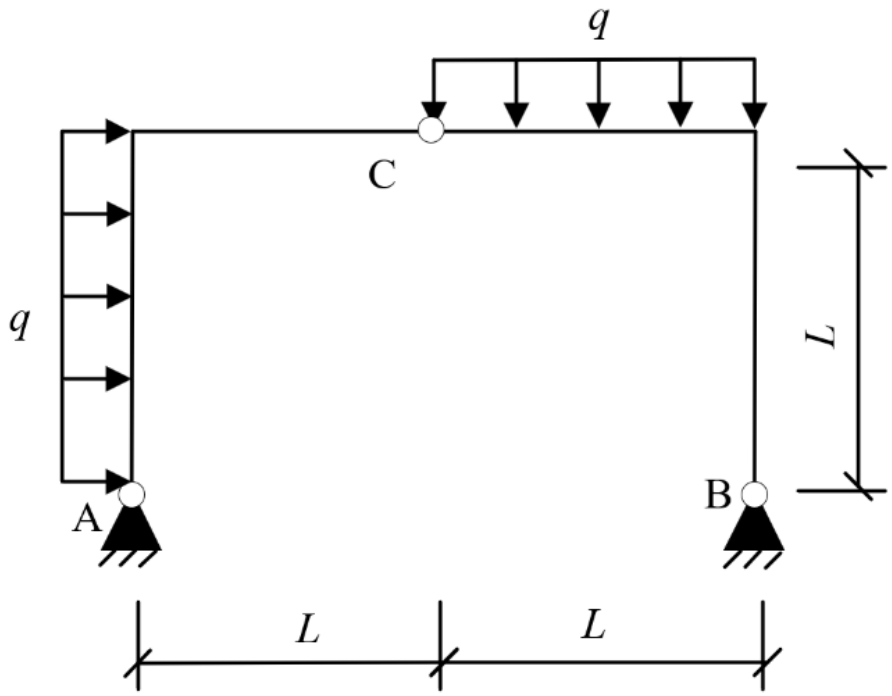


Figure 1(b). Plane frame

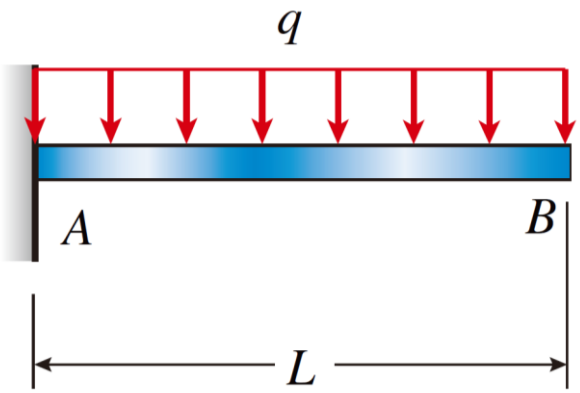


Figure 2(a). Cantilever beam

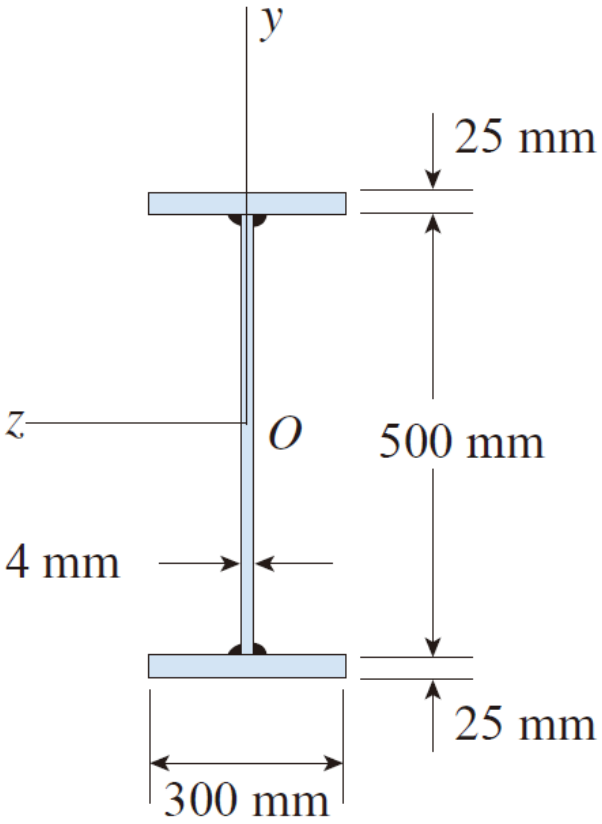


Figure 2(b). Beam section

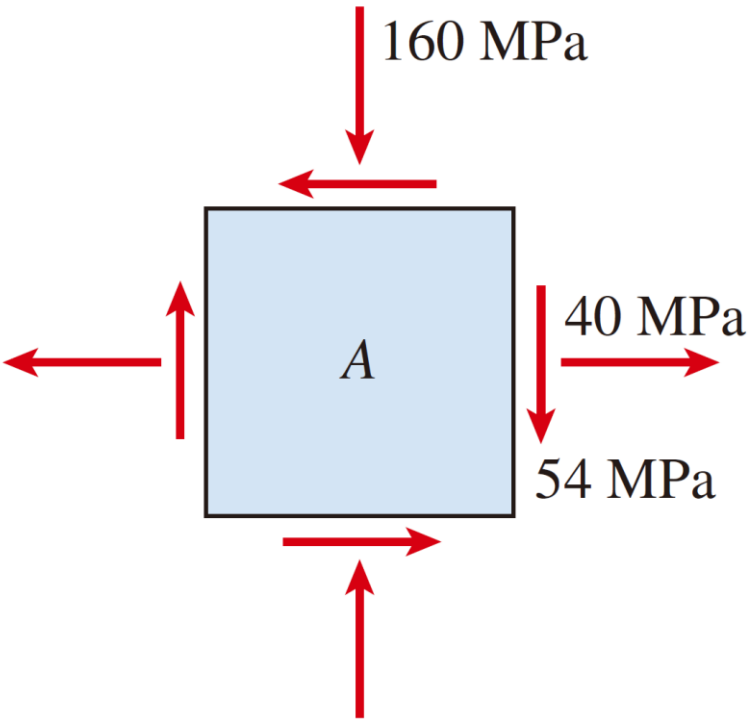


Figure 2(c). Plane stress at a point

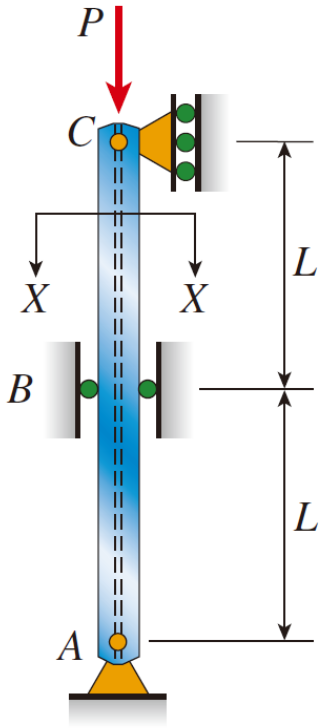


Figure 3a. Column

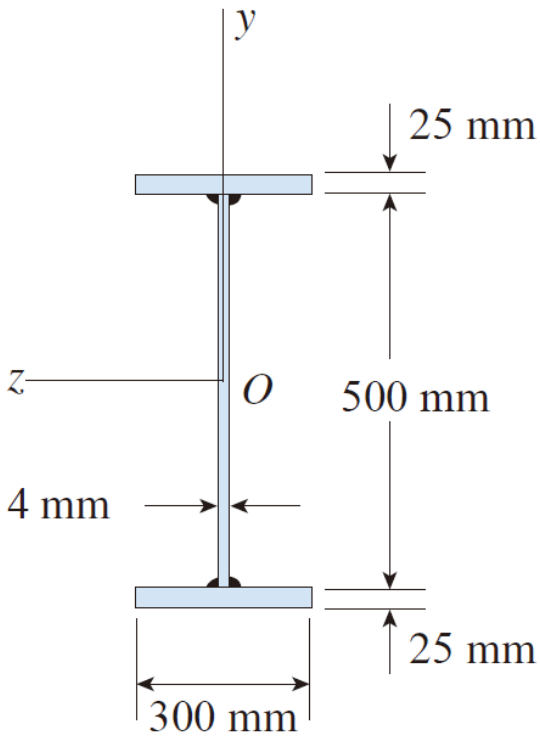


Figure 3b. Section X-X

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